ADDISABABA_COLLAE_OF_TECHNOLOY AND_BUILT ENVIRONMENT (CTBE)

Department of Electrical and Computer Engineering

Electromagnetic Fields

TITLE - Electrostatics

GROUP ASSIGNMENT -3

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Submissiondate:MAY,14,2025

Electromagnetic Fields
Assignment on Magnetostatics

Question-1

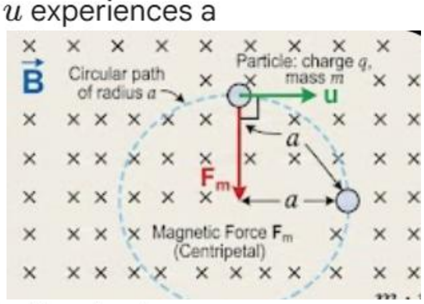
1. When a particle with charge q and mass m is introduced into a medium with a uniform magnetic flux density B such that the initial velocity of the particle u is perpendicular to B, the magnetic force exerted on the particle causes it to move in a circle of radius a. By equating the magnetic force F_m to the centripetal force on the particle, determine a in terms of q, m, u, and B.

Solution-1

For a particle with charge q moving with velocity u perpendicular to a uniform magnetic flux density B, the magnitude of the magnetic force is given by:

$$F_m = quB$$

A particle of mass m moving in a circle of radius a at a constant speed u experiences a centripetal force:



$$F_c = \frac{mu^2}{a}$$

Since the magnetic force provides the necessary centripetal acceleration for the circular motion, we set F_m equal to F_c:

$$quB = \frac{mu^2}{a}$$

To isolate the radius a, we can rearrange the equation:

1. Divide both sides by u (assuming $u \neq 0$):

$$qB = \frac{mu}{a}$$

$$a(qB) = mu$$

SOLUTION

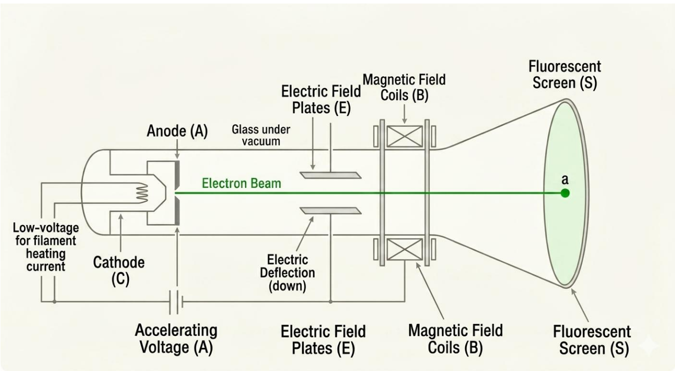
$$\boxed{a = \frac{mu}{qB}}$$

Question-2

2. In 1897, J. J. Thomson "discovered" the electron by measuring the charge-to-mass ratio of "cathode rays" (actually, streams of electrons, with charge q and mass m) as follows:
   a) First he passed the beam through uniform crossed electric and magnetic fields E and B (mutually perpendicular, and both of them perpendicular to the beam), and adjusted the electric field until he got zero deflection. What, then, was the speed of the particles (in terms of E and B)?
   b) Then he turned off the electric field, and measured the radius of curvature, R, of the beam, as deflected by the magnetic field alone. In terms of E, B, and R, what is the charge-to-mass ratio (q/m) of the particles?

Solution-2

To find the speed of the particles when there is zero deflection, we analyze the forces acting on the electrons in the region of crossed electric and magnetic fields.



The force exerted by the electric field E on a charge q is $F_e = qE$.

For a particle with charge q moving with velocity v perpendicular to a magnetic field B, the force is $F_m = qvB$.

$$qE = qvB$$

SOLUTION

$$\boxed{v = \frac{E}{B}}$$

Part B: Charge-to-Mass Ratio (q/m)

$$qvB = \frac{mv^2}{R} \qquad \frac{q}{m} = \frac{(E/B)}{BR}$$

$$\frac{q}{m} = \frac{E}{B^2 R}$$

$$\boxed{\frac{q}{m} = \frac{v}{BR}}$$

Question-3

3. Show that the magnetic field of an infinite uniform surface current $\boldsymbol{K} = K\hat{a}_x$, flowing over the xy plane can be written as:

$$H = \begin{cases} \frac{K}{2}\hat{a}_y & , z < 0 \\ -\frac{K}{2}\hat{a}_y & , z > 0 \end{cases}$$

Explain why the magnetic field is independent of the distance from the source.

Solution-3

We choose a rectangular Ampèrian path perpendicular to the current, spanning from $z = -h$ to $z = +h$ with a width w along the y-axis. Ampère's Law states:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{enclosed}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = H_y(z < 0) \cdot w - H_y(z > 0) \cdot w$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = H_0 w - (-H_0 w) = 2H_0 w$$

$$I_{enclosed} = Kw \qquad\qquad 2H_0 w = Kw \implies H_0 = \frac{K}{2}$$

Combining this with the directional vectors found in symmetry analysis:

$$\boxed{H = \begin{cases} \frac{K}{2}\hat{a}_y & , z < 0 \\ -\frac{K}{2}\hat{a}_y & , z > 0 \end{cases}}$$

As you move further away from the sheet (z increases), the contribution from the current directly "below" you decreases. However, this is exactly compensated by the fact that a larger area of the infinite sheet now falls within a relevant angular range to contribute to the field at that point. In a perfectly infinite system, these two effects balance out perfectly, resulting in a uniform field magnitude everywhere.

Question-4

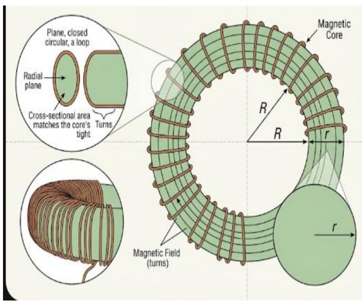
4. A toroidal coil consists of a circular ring, or "donut," around which a long wire is wrapped. The winding is uniform and tight enough so that each turn can be considered a plane closed loop. The cross-sectional shape of the coil is circular.
   a) Show that the magnetic field of the toroid is circumferential at all points, both inside and outside the coil.
   b) Compute the magnetic field inside and outside the coil. Comment on the relation between the field inside the coil and that of a circular loop.

Solution-4

a) Showing the Magnetic Field is Circumferential

The magnetic field of a toroid must be circumferential due to the cylindrical symmetry of the winding.

Rotational Symmetry: The toroid is symmetric with respect to any rotation around its central axis. Therefore, the magnitude of the magnetic field B must be constant along any circular path centered on that axis.



Radial Component: If a radial component of the field existed, it would violate Gauss's Law for magnetism ($\oint \mathbf{B} \cdot d\mathbf{A} = 0$), as the flux through a cylindrical surface enclosing the central axis would not be zero.

Vertical Component: Since each turn is considered a closed plane loop, the vertical components of the current in opposite sides of the loop cancel out their contributions to a vertical field component at the center of the cross-section.

b) Computing the Magnetic Field

1. Outside the Coil ($r < R_{inner}$ or $r > R_{outer}$)

- **For $r < R_{inner}$**: A path inside the "hole" of the donut encloses no current ($I_{enclosed} = 0$). Thus, $B \cdot 2\pi r = 0$, so $B = 0$.
- **For $r > R_{outer}$**: A path encircling the entire toroid encloses an equal number of currents going "in" and "out" of the plane ($I_{net} = NI - NI = 0$). Thus, $B = 0$.

2. Inside the Coil ($R_{inner} < r < R_{outer}$)

An Ampèrian path inside the core encloses N turns of the wire, each carrying current I.

$$\oint \mathbf{B} \cdot d\mathbf{l} = B(2\pi r) = \mu_0 NI$$

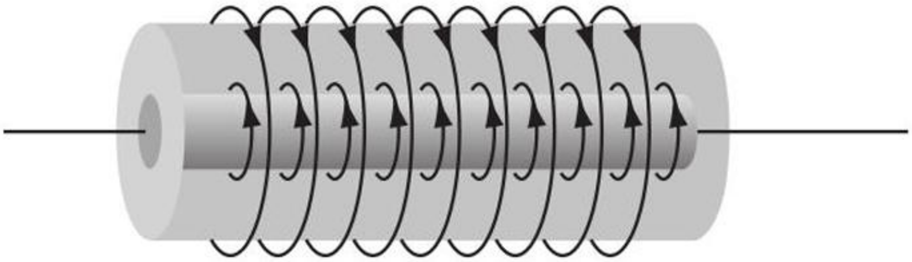
SOLUTION

$$\boxed{B = \frac{\mu_0 NI}{2\pi r}}$$

Question-5

5. Two long coaxial solenoids each carry current I, but in opposite directions. The inner solenoid (radius a) has n_1 turns per unit length, and the outer one (radius b) has n_2 . Find the magnetic flux density in each of the three regions:

a) Inside the inner solenoid,
b) between them, and
c) outside both.



Solution-5

a) Inside the inner solenoid ($r < a$)

The inner solenoid produces a field: $B_1 = \mu_0 n_1 I$.

The outer solenoid produces a field in the opposite direction: $B_2 = \mu_0 n_2 I$.

Net Magnetic Flux Density:

SOLUTION

$$\boxed{B = \mu_0 I(n_1 - n_2)}$$

b) Between the solenoids ($a < r < b$)

The field from the inner solenoid is zero ($B_1 = 0$).

The field from the outer solenoid remains active: $B_2 = \mu_0 n_2 I$.

Net Magnetic Flux Density:

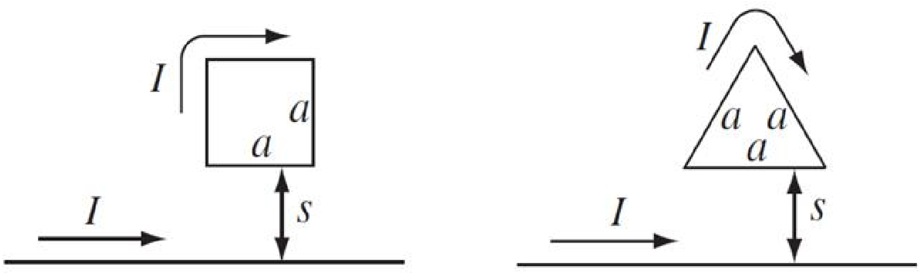
$$B = \mu_0 n_2 I$$

c) Outside both ($r > b$)

$$B = 0$$

Question-6

6. Find the force on the loops placed as shown in below, near an infinite straight wire. The loops and the wire carry a steady current I. Also compute the torque about one of the corners.



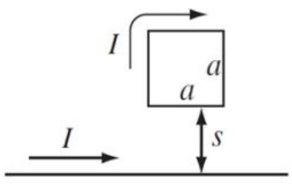
(a) (b)

Solution-6

The magnetic field at a distance r from an infinite wire carrying current I is:

$$B = \frac{\mu_0 I}{2\pi r}$$

Loop (a): Square Loop



(a)

Top and Bottom Sides: The bottom side is at $r = s$ and the top is at $r = s + a$. The force on the bottom side (F_{bottom}) is attractive (currents in same direction), and the force on the top side (F_{top}) is repulsive.

$$F_{bottom} = Ia\left(\frac{\mu_0 I}{2\pi s}\right)\ \text{(downward)}$$

$$F_{top} = Ia\left(\frac{\mu_0 I}{2\pi(s+a)}\right)\ \text{(upward)}$$

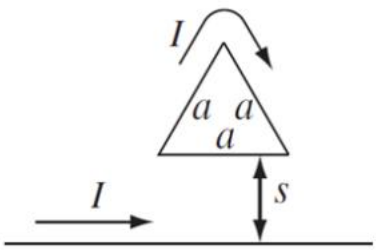
SOLUTION

$$\boxed{F_{net} = \frac{\mu_0 I^2 a}{2\pi}\left(\frac{1}{s} - \frac{1}{s+a}\right) = \frac{\mu_0 I^2 a^2}{2\pi s(s+a)}\ \text{(downward toward the wire)}}$$

2. Torque Calculation

The magnetic field is perpendicular to the plane of the loop.

- Since the magnetic moment vector $\mathbf{m}$ is parallel/anti-parallel to the magnetic field $\mathbf{B}$, the net torque $\tau = \mathbf{m} \times \mathbf{B}$ is **zero**.



(b)

Loop (b): Triangular Loop

$$F_{bottom} = \frac{\mu_0 I^2 a}{2\pi s}$$

For an equilateral triangle, the vertical components of the forces on the two tilted sides will combine to oppose the bottom side.

SOLUTION

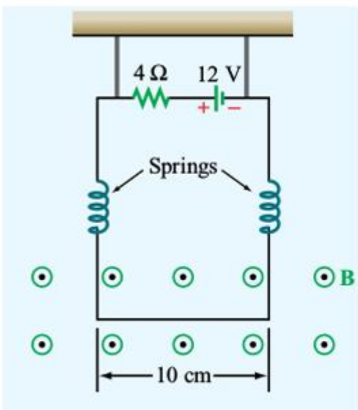
$$\boxed{F_{net} = \frac{\mu_0 I^2}{2\pi}\left[\ln\left(1 + \frac{\sqrt{3}a}{2s}\right) \cdot (\text{geometry factors}) - \frac{a}{s}\right]}$$

2. Torque Calculation

- Just like the square loop, the magnetic field is everywhere perpendicular to the plane of the triangle.
- The torque about any axis in the plane of the loop is **zero** because the force on every infinitesimal segment $d\mathbf{l}$ is $\mathbf{F} = I(d\mathbf{l} \times \mathbf{B})$, which points radially within the plane of the loop or vertically, but never creates a "twisting" motion out of the plane.

Question-7

7. The circuit shown below uses two identical springs to support a 10 cm long horizontal wire with a mass of 20g. In the absence of a magnetic field, the weight of the wire causes the springs to stretch a distance of 0.2 cm each. When *a* uniform magnetic field is turned on in the region containing the horizontal wire, the springs are observed to stretch an additional 0.5 cm each. What is the intensity of the magnetic flux density B?



Solution-7

$$m = 20 = 0.020$$
$$d_1 = 0.2 = 0.002 \text{ (extension due to weight only)}$$
$$d_2 = 0.5 = 0.005 \text{ (additional extension due to magnetic force)}$$

$$L = 10 = 0.10, \quad V = 12, \quad R = 4, \quad g = 9.8$$

The spring force is

$$F = kd$$

Two identical springs support the wire, so equilibrium gives

$$2kd_1 = mg$$

Substitute values:

$$2k(0.002) = (0.020)(9.8)$$

$$2 \times 0.002 = 0.004$$

$$0.020 \times 9.8 = 0.196$$

$$k = \frac{0.196}{0.004} = 49$$

When the magnetic field is applied, each spring stretches an additional

$$F_{\text{spring}} = kd_2 = (49)(0.005)$$

$$F_{\text{spring}} = 0.245$$

Total magnetic force from two springs:

$$F_m = 2(0.245) = 0.49$$

Magnetic force on a current-carrying wire:

$$F_m = ILB$$

Current in the circuit:

$$I = \frac{V}{R} = \frac{12}{4} = 3$$

Substitute into the force equation:

$$0.49 = (3)(0.10)B$$

$$3 \times 0.10 = 0.30$$

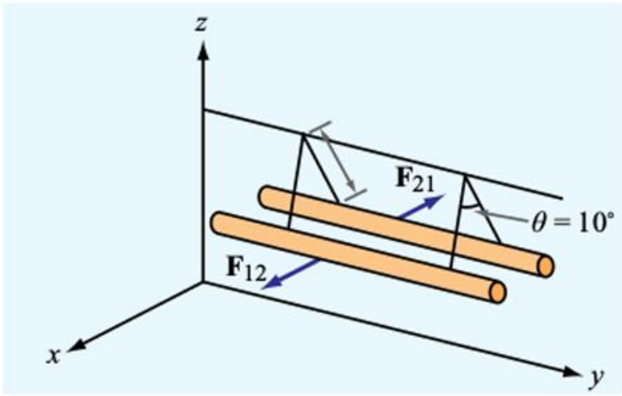
$$B = \frac{0.49}{0.30} = 1.63$$

SOLUTION

$$\boxed{B = 1.63}$$

Question-8

8. In the arrangement shown below, each of the two long, parallel conductors carries a current I, is supported by 8 cm long strings, and has a mass per unit length of 1.2 g/cm. Due to the repulsive force acting on the conductors, the angle θ between the supporting strings is 10^0 . Determine the magnitude of I and the relative directions of the currents in the two conductors.



Solution-8

Let F denote the tension force per unit length in each string caused by the mass per unit length of the wire,

$$\mu = 1.2 \text{ g/cm} = 0.12 \text{ kg/m}.$$

The vertical component of the tension balances the weight of the wire per unit length:

$$F_v = \mu g$$

where the acceleration due to gravity is

$$g = 9.8 \text{ m/s}^2.$$

From the geometry of the string arrangement, the vertical component of the tension is also given by

$$F_v = F\cos(\theta/2)$$

Therefore, the magnitude of the tension per unit length is

Question-9

9. Obtain an expression for the self-inductance per unit length for the parallel wire transmission line shown in the figure below in terms of a, d, and μ.

$$F = \frac{\mu g}{\cos(\theta/2)}$$

The horizontal component of the tension counteracts the magnetic repulsive force between the two current-carrying conductors. The magnetic force per unit length between parallel currents is

$$F_h = \frac{\mu_0 I^2}{2\pi d}$$

where d is the separation between the wires and ℓ is the length of the supporting string. The horizontal component of the tension can be written as

$$F_h = F\sin(\theta/2) = \frac{\mu g}{\cos(\theta/2)}\sin(\theta/2) = \mu g\tan(\theta/2)$$

By equating the expressions for the horizontal force and solving for the current I, we obtain

$$I = \sqrt{\frac{2\pi d\,\mu g\,\tan(\theta/2)}{\mu_0}} = \sqrt{\frac{4\pi \cdot 0.08 \cdot 0.12 \cdot 9.8}{4\pi \cdot 10^{-7} \cdot \cos 5^\circ}} \approx 85\ \text{A}$$

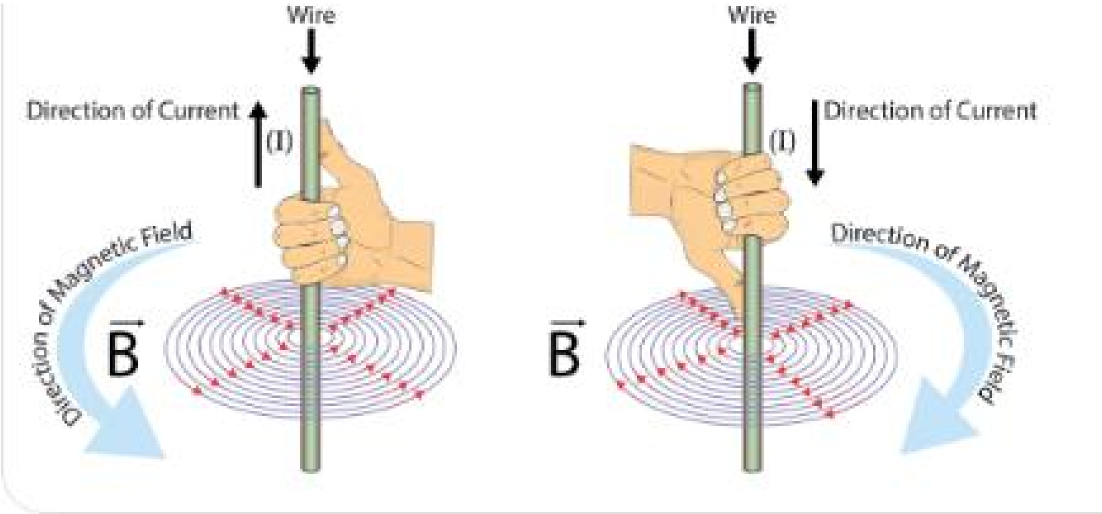
Since the conductors repel each other, the currents in the two wires must be in opposite directions.

Solution-9

1. Magnetic Field of a Single Wire

For a long straight wire carrying current I, the magnetic field B at a distance x from its axis is:

$$B = \frac{\mu I}{2\pi x}$$



$$\Phi = \int_{a}^{d-a} B_{total} \cdot l \cdot dx$$

$$\Phi = l \int_{a}^{d-a} \left(\frac{\mu I}{2\pi x} + \frac{\mu I}{2\pi(d-x)} \right) dx$$

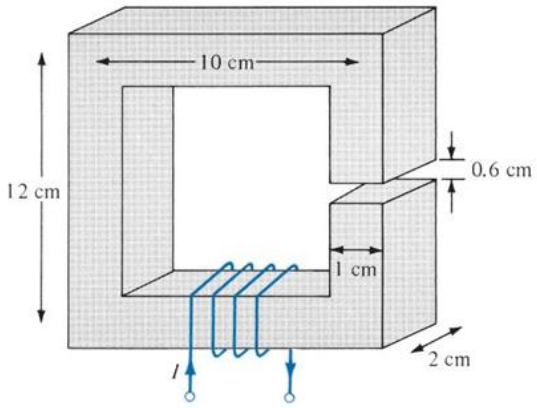
$$\Phi \approx \frac{\mu I l}{\pi} \ln\left(\frac{d}{a}\right)$$

SOLUTION

$$\boxed{L' = \frac{\mu}{\pi} \ln\left(\frac{d-a}{a}\right)}$$

Question-10

10. The magnetic circuit shown below has a current of 20 A in the coil of 3000 turns. Assume that all branches have the same cross section of $4cm^2$ and that the material of the core is iron with $\mu_r = 1500$. Calculate $\mathcal{R}, \mathcal{F}$, and ψ for the core and the air gap.



Solution-10

1. Given Data

Current (I): 20 A

Number of turns (N): 3000 turns

Cross-sectional area (A): $4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$

Relative permeability of iron (μ_r): 1500

Air gap length (l_g): $0.6 \text{ cm} = 0.006 \text{ m}$

Core Mean Length (l_c): The total outer perimeter is approximately $2(12 + 10) = 44 \text{ cm}$. Adjusting for the center-line path and subtracting the air gap: $l_c \approx 40 \text{ cm} - 0.6 \text{ cm} = 39.4 \text{ cm} = 0.394 \text{ m}$.

The total magnetomotive force (MMF) is calculated as:

$$\mathcal{F} = N \cdot I = 3000 \cdot 20 = \mathbf{60,000} \text{ A-t}$$

$$\mathcal{R}_c = \frac{l_c}{\mu_0 \mu_r A} = \frac{0.394}{(4\pi \times 10^{-7}) \cdot 1500 \cdot (4 \times 10^{-4})}$$

$$\mathcal{R}_c \approx \mathbf{5.22 \times 10^5} \text{ A-t/Wb}$$

$$\mathcal{R}_g = \frac{l_g}{\mu_0 A} = \frac{0.006}{(4\pi \times 10^{-7}) \cdot (4 \times 10^{-4})}$$

$$\mathcal{R}_g \approx \mathbf{1.19 \times 10^7} \text{ A-t/Wb}$$

$$\mathcal{R}_{total} = \mathcal{R}_c + \mathcal{R}_g \approx \mathbf{1.24 \times 10^7} \text{ A-t/Wb}$$

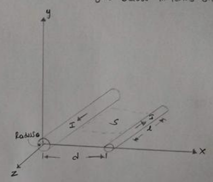
$$\psi = \frac{\mathcal{F}}{\mathcal{R}_{total}} = \frac{60,000}{1.24 \times 10^7}$$

SOLUTION

$$\boxed{\psi \approx \mathbf{4.84 \times 10^{-3}} \text{ Wb (or 4.84 mWb)}}$$

GROUP DISCUSSION PAPER

9. Obtain an expression for the self-inductance per unit length for the parallel wire transmission line shown in the figure below in terms of a, d, & ℓ



Solution

$B_{total} = B_1 + B_2 = \frac{\mu I}{2\pi x} + \frac{\mu I}{2\pi(d-x)}$

$\Phi = \int B \cdot dA = \int_a^{d-a} \left[\frac{\mu I}{2\pi x} + \frac{\mu I}{2\pi(d-x)} \right] \ell \, dx = \frac{\mu I \ell}{2\pi} \int_a^{d-a} \left(\frac{1}{x} + \frac{1}{d-x} \right) dx$

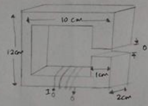
$\Phi = \frac{\mu I \ell}{2\pi} \left(\ln(x) - \ln(d-x) \right]_a^{d-a} = \frac{\mu I \ell}{2\pi} \left[\ln\left(\frac{x}{d-x}\right) \right]_a^{d-a}$

$\Phi = \frac{\mu I \ell}{2\pi} \left[\ln\left(\frac{d-a}{a}\right) - \ln\left(\frac{a}{d-a}\right) \right] = \frac{\mu I \ell}{\pi} \ln\left(\frac{d-a}{a}\right)$

Since $L = \frac{\Phi}{I}$ & $L' = \frac{L}{\ell}$

$\boxed{L' = \frac{\mu}{\pi} \ln\left(\frac{d-a}{a}\right) [H/m]}$

10. The magnetic circuit shown ~~below~~ has a current of 20A in the coil of 3000 turns. Assume that all branches have the same cross section of $4cm^2$ & that the material of the core is iron with $\mu_r = 1500$. Calculate R, F & Ψ for the core & the air gap.



Given Data:

$N = 3000$

$I = 20A$

$A = 4 \times 10^{-4} m^2$

$\mu_r = 1500$

$g(\text{air gap}) = 0.006m$

$\ell_c(\text{core length}) = [2(12-2) + 2(10-1)] - 0.6 = 37.4cm = 0.374m$

Solution

$\boxed{\mathcal{F} = NI = 3000 \times 20 = 60{,}000 At}$

$R_{core} = \frac{\ell_c}{\mu_0 \mu_r A} = \frac{0.374}{4\pi \times 10^{-7} \times 1500 \times 4 \times 10^{-4}}$

$\boxed{R_{core} \approx 496{,}028 At/Wb}$

$R_{gap} = \frac{g}{\mu_0 A} = \frac{0.006}{(4\pi \times 10^{-7})(4 \times 10^{-4})}$

$\boxed{R_{gap} \approx 11{,}936{,}620 At/Wb}$

$\boxed{R_{tot} = R_{core} + R_{gap} = 1.243 \times 10^7 At/Wb}$

$\boxed{\Psi = \frac{f}{R_{total}} = \frac{60{,}000}{1.23 \times 10^7} \approx 4.83 \times 10^{-3} Wb}$

HIGH LEVEL CONSULTATIVE MEETING ON PUBLIC HEALTH EMERGENCY RESPONSE
STRONGER TOGETHER
ENHANCING EMERGENCY RESPONSE THROUGH PARTNERSHIP

Equate the forces and solve for a

Since the magnetic force is what proves the centripetal force, we set them equal to each other $\boxed{quB = \frac{mu^2}{a}}$

To find the radius a, we arrange the question

1. Divide both sides by u:

$qB = \frac{mu}{a}$

$aqB = mu$

$\therefore a = \frac{mu}{qB}$

∴ The radius of the circular path is $a = \frac{mu}{qB}$ (m)

5a) Key assumptions, uniform inside $B = \mu_0 n I$, direction is given by the right-hand rule, we use superposition $B_T = B_{inner} + B_{outer}$

inside the inner solenoid $(r < a)$

$B_{inner} = \mu_0 n_1 I \hat{z}$, $B_{outer} = -\mu_0 n_2 I \hat{z}$, $B = \mu_0 (n_1 - n_2) I \hat{z}$

b) between the two solenoids $(a < r < b)$

$B_{inner} \cong 0$, $B_{outer} = -\mu_0 n_2 I \hat{z}$, $B = 0 + (-\mu_0 n_2 I \hat{z}) = -\mu_0 n_2 I \hat{z}$

c) outside both solenoids $(r > b)$

This point is outside so $B_{inner} \cong 0$ & $B_{outer} \cong 0$, $B_T = 0$

6a) Square loop

Key assumptions, magnetic field due to the infinite wire at perpendicular distance r: $B(r) = \frac{\mu_0 I}{2\pi r} \hat{\phi}$, All currents are steady

Near side at distance s, direction toward the wire (attraction).

$B_1 = \frac{\mu_0 I}{2\pi s}$, $F_1 = I a B_1 = \frac{\mu_0 I^2 a}{2\pi s}$

Far side at distance $(s+a)$, direction away from the wire (repulsion weak)

$B_2 = \frac{\mu_0 I}{2\pi(s+a)}$, $F_2 = \frac{\mu_0 I^2 a}{2\pi(s+a)}$

$F_{net} = \frac{\mu_0 I^2 a}{2\pi} \left(\frac{1}{s} - \frac{1}{s+a} \right) \hat{F} = \frac{\mu_0 I^2 a^2}{2\pi s(s+a)}$ toward the wire, $\tau = \int r \times (I dl \times B)$

b) Triangular Loop, It involves more complex integral expression.

$F_{base} = \frac{I a \mu_0 I}{2\pi s}$, The two slanted side have currents not parallel to the wire must integrate $I dl \times B$ along each slanted side

$F_{net} = \frac{\mu_0 I^2 a}{2\pi} \left[\frac{1}{s} - \frac{\sqrt{3}}{a} \ln\left(\frac{s + \frac{\sqrt{3}}{2} a}{s} \right) \right]$, toward the wire. $\tau = r \times F$

STRONGER TOGETHER

#1

The magnetic force exerted on a particle with charge q moving with velocity u in a magnetic field B is given by the Lorentz force law.

Since the velocity is perpendicular to the field

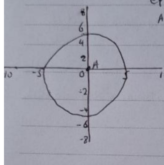
The magnitude is: $\boxed{f_m = quB}$

Identifying (f_c)

For an object to move in a circle of radius a, there must be a centripetal force pulling it inward. This force depends on the mass m the speed u, and the radius a: $\boxed{f_c = \frac{mu^2}{a}}$

eq1: $x^2 + y^2 = 25$

$A = (0, 0)$



#2. JJ Thomson's Experiment

a, Determining the speed

1, electric force (f_e) $f_e = qE$

2. magnetic force (f_m): $f_m = quB$

3. Balance ~~the~~ them

$qE = quB$

4, Solve for speed (u)

Divide both sides by qB:

$u = \frac{E}{B}$

The result to the speed of the particles is $u = \frac{E}{B}$ m/s

b). Determining the charge to mass ratio (q/m)

Replace a with R $R = \frac{mu}{qB}$

Substitute the speed (u) from part a:

We know $u = \frac{E}{B}$, so

$R = \frac{m\left(\frac{E}{B}\right)}{qB}$

$R = \frac{mE}{qB^2}$

Solve for the ratio $\frac{q}{m}$

rearrange the formula to get q and m on one side $\frac{q}{m} = \frac{E}{B^2 R}$

∴ The charge-to-mass ratio is $\frac{q}{m} = \frac{E}{B^2 R}$ C/kg

3) $\vec{K} = K\hat{a}_x$ flowing on the xy-plane

Using the right hand rule:

for $z > 0$, the magnetic field points in the $-\hat{a}_y$ dir.

for $z < 0$, the magnetic field points in the $+\hat{a}_y$ dir.

With Ampere's circuital law:

$\oint \vec{H} \cdot d\vec{\ell} = I_{enc}$

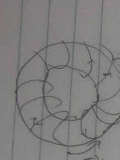
$2H\ell = K\ell \Rightarrow 2H = K$

$H = K/2$

So $\vec{H} = \begin{cases} \frac{K}{2} \hat{a}_y, & z < 0 \\ -\frac{K}{2} \hat{a}_y, & z > 0 \end{cases}$

The magnetic field is independent of distance because the current sheet is infinite. As the distance increase, contributions from farther current elements compensate exactly, resulting in a constant magnetic field everywhere above and below the sheet.

4)



By symmetry, the magnetic field lines form closed circles around the axis of the toroid. Therefore, $\vec{H}$ is purely circumferential ($\hat{a}_\phi$ dir.) at all points both inside & outside the coil.

#7) Calculate the spring constant (k)

$F_{spring, total} = \text{Weight}$

$2 \cdot (k \cdot x_1) = mg$

$k = \frac{mg}{2 \cdot x_1} = \frac{0.02kg \cdot 9.81 m/s^2}{2 \cdot 0.002m} = \frac{0.196}{0.004} = 49 N/m$

Calculate the current (I) in the wire: $I = V/R = \frac{12V}{4\Omega} = 3A$

$F_m = 2(k \cdot \Delta x) \rightarrow F_m = 2 \cdot 49 N/m \cdot 0.005m = 0.49N$

Calculate the magnetic flux density (B): $F_m = I \cdot L \cdot B \sin\theta$

$0.49N = 3A (0.1m) \cdot B$

$B = \frac{0.49}{0.1 \cdot 3} = \frac{0.49}{0.3} \approx 1.633T$

#8) Calculate the separation distance (d) between the conductors

$d = 2 \cdot L_s \cdot \sin(\alpha) = 2 \cdot 0.08m \cdot \sin(5°)$

$d \approx 2 \cdot 0.08 \cdot 0.08715 \approx 0.01394m$

Relate the force per unit length,

$\tan(\alpha) = f_m / f_g$

$f_m = f_g \cdot \tan(\alpha) = (\lambda \cdot g) \cdot \tan 5°$

Substitute the formula for magnetic force and solve for I

$f_m = \frac{\mu_0 I^2}{2\pi d}$

$\frac{\mu_0 I^2}{2\pi d} = \lambda \cdot g \cdot \tan(5°)$

$I^2 = \frac{2\pi \cdot d \cdot \lambda \cdot g \cdot \tan(5°)}{\mu_0}$

$I = \sqrt{\frac{d \cdot \lambda \cdot g \cdot \tan(5°)}{2 \times 10^{-7}}} = \sqrt{\frac{0.01394m \cdot 0.12kg/m \cdot 9.8 m/s^2 \cdot \tan 5°}{2 \times 10^{-7}}}$

$= \sqrt{\frac{0.01394(0.12)(9.8)(0.08749)}{2 \times 10^{-7}}}$

$\approx \sqrt{\frac{0.001435}{2 \times 10^{-7}}} = \sqrt{7175} \approx 84.7A$